

native compiler

↳ high level code to native machine instructions

cross-compiler

↳ high level to executable that other platforms can run

transpiler

↳ input high-level to some other target high-level

decompiler [difficult due to loss of info]

↳ low level to high level

JIT compiler (eg: Java)

↳ runs after program started

↳ compiles code "just-in-time"

↳ ahead-of-time compiler + interpreter

self compiling compiler

↳ compiles itself → written in same language it compiles

↳ Bootstrapping: how its built.

small basic compiler in existing lang / by hand

→ compiles a more advanced version written for new lang

n iterations

→ ... → compiles full compiler in new lang

compiler-compiler

↳ takes old compiler and generate new one

→ input: grammar
output: parser code

→ typically only deals with syntax analysis

incremental compiler

↳ compiles only changes